Sahil Kumar

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Education

Indian Institute of Technology (IIT) Madras — Chennai, India
Bachelor of Science (B.S.) in Data Science and Applications — May 2024 – Dec 2028
CGPA: 7.84/10 *Relevant Coursework: Machine Learning Techniques, Statistical Learning, Deep Learning, NLP*

National Institute of Technology (NIT) Uttarakhand — Srinagar, India
Bachelor of Technology (B.Tech) in Electronics and Communication Engineering — Aug 2023 – May 2027
CGPA: 8.55/10 *Relevant Coursework: Probability & Statistics, Vector Calculus, Linear Algebra, Matrix Decompositions*

Experience

Accenture — India
Custom Software Engineering Associate Intern (GenAI Integration) — May 2026 – Present

- Architected production-grade **Retrieval-Augmented Generation (RAG)** pipelines using **LangChain** and **ChromaDB** over enterprise document corpora, optimizing vector-based semantic search workflows.
- Integrated **Large Language Models (LLMs)** into multi-tier applications utilizing **Amazon Bedrock** and **Hugging Face APIs**; applied prompt engineering strategies to improve inference accuracy.
- Developed and deployed high-throughput **FastAPI** microservices using **Docker** and **AWS** infrastructure within an Agile ecosystem to support concurrent production AI workloads.

Projects

Toxic Comment Classification using NLP & Ensemble Learning — [GitHub]
Python, Scikit-Learn, LightGBM, TF-IDF, StratifiedKFold, Nelder-Mead Threshold Tuning

- Engineered a multi-class toxicity detection system on **198K+ instances**, achieving an **81.87% Macro F1-score** on a heavily imbalanced dataset (minority class at 2.8%).
- Constructed a **35K-dimensional sparse feature matrix** combining character/word-level **TF-IDF** with 55 dense linguistic features, text-engagement metrics, and cross-interaction tokens.
- Optimized and blended **LightGBM** and **Logistic Regression** architectures using **Nelder-Mead threshold tuning** and 16× custom class weighting via 3-fold stratified cross-validation, achieving a **+6.4pp gain** over SVM baselines.

Multimodal Price Prediction System | Amazon ML Challenge — [GitHub]
PyTorch, CLIP, LoRA, QLoRA, Hugging Face, PEFT, Transformers, Distributed Training

- Built a multimodal deep learning pipeline in **PyTorch** fusing **CLIP** vision-language representations with structured product metadata to predict consumer price scales across large-scale e-commerce datasets.
- Implemented parameter-efficient fine-tuning (PEFT) using **LoRA** and **QLoRA** via Hugging Face to optimize distributed training and minimize hardware overhead while retaining competitive convergence rates.

Technical Skills

Languages: Python, SQL, C,
Machine Learning: Scikit-Learn, Feature Engineering, Ensemble Learning (LightGBM), Stratified Cross-Validation, Class Imbalance Handling, Nelder-Mead Optimization
Deep Learning & NLP: PyTorch, Transformers, LLMs, CLIP, LoRA, QLoRA, PEFT, Hugging Face, Attention Mechanisms
Generative AI: Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (ChromaDB, FAISS), Semantic Search, Prompt Engineering, Context Window Management
Math & Tools: Probability & Statistics, Linear Algebra, Vector Calculus, Docker, AWS (Amazon Bedrock), Git, Weights & Biases (W&B)

Achievements & Leadership

- **Student Placement Representative**, NIT Uttarakhand — Elected to manage corporate relations and coordinate industrial outreach.
- **Top Tier Performer**, IIT Madras Comment Category Prediction Challenge — Secured **81.87% Macro F1-score** on 198K multi-class samples using custom ensembling.
- Solved **150+ Data Structures & Algorithms (DSA) problems** on LeetCode covering dynamic programming, graphs, and tree structures.